

CP Generator

Algorithmic Methods for Origami Crease Patterns

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What this project is

A desktop app that generates, optimizes, and animates a flat-foldable crease pattern on a square sheet — entirely from scratch.

`cp.py · cp_gen.py · fold_sim.py`

Four linked algorithms

1. Build a planar candidate graph.
2. Repair parity & optimize for Kawasaki.
3. Assign mountains/valleys.
4. Extract faces & animate the fold.

Key objects

- Sheet $[0, s]^2$; graph $G = (V, E)$, positions $X_v \in \mathbb{R}^2$.
- Crease label $\text{type}(e) \in \{-1, 0, 1\}$: unassigned / mountain / valley.
- **Sector angle**: angle between consecutive creases at a vertex.
- **Half-edge**: oriented edge — determines which face lies to its left.

Algorithm I — Planar candidate graph

Goal. A planar graph dense enough for interesting folds, regular enough for stable local angle computation.

Construction

1. Sample interior points; add four corners.
2. Snap near-boundary points to the boundary.
3. Compute the Delaunay triangulation.
4. Use its edges as the initial crease graph.

Why Delaunay?

No other input point lies inside the circumcircle of any triangle — avoids skinny triangles and gives clean local angle data.

Algorithm II — Parity repair & Kawasaki optimization

Necessary condition

Every interior vertex must have **even degree**: $d(v) \equiv 0 \pmod{2}$.

Repair. Pair odd-degree vertices; delete the shortest path between each pair. (Heuristic — not globally optimal.)

Geometry repair. Let $D(X) = \sum_v w_v \|X_v - X_{0,v}\|^2$. Move vertices as little as possible, subject to the Kawasaki constraints:

$$\min_X D(X) + \frac{\alpha}{2} D(X)^2,$$

Boundary weights w_v are large; interior weights are small. Solved with SLSQP.

Kawasaki's theorem (local)

Sector angles $\alpha_1, \dots, \alpha_{2m}$ must satisfy

$$\alpha_1 + \alpha_3 + \dots = \alpha_2 + \alpha_4 + \dots = \pi.$$

Algorithm III — Local constraints: Maekawa & Bern–Hayes

Maekawa's theorem (local)

At every interior vertex,

$$M(v) - V(v) = \pm 2.$$

Used as an **admissibility filter**: reject any assignment violating this anywhere.

Bern–Hayes-style reduction

For each vertex, repeatedly **remove the smallest sector**: its two bounding creases are forced *equal* or *opposite* in sign.

1. Sort creases around the vertex by angle.
2. Find the smallest wedge; record its forced sign relation.
3. Recurse until the vertex is fully resolved.

Boundary vertices get temporary virtual rays to close the cyclic order.

Algorithm III — Merging, exact search, & interlacing

Each vertex reduction yields sign chains:

$$\sigma(e_i) = \sigma(e_j) \quad \text{or} \quad \sigma(e_i) = -\sigma(e_j).$$

Exact search

- Merge chains across shared creases into g free groups.
- Enumerate all 2^g seed assignments; propagate implied signs.
- Keep only assignments that pass Maekawa at every vertex.

Undetermined creases keep label -1 ; no sign is invented.

Interlacing preference

Among admissible assignments, prefer $M, V, M, V, \dots$ over $M, M, V, V, \dots$; weight sharper wedges more.

Does not affect correctness.

Algorithm IV — Faces, reflection & 3D animation

Face extraction

Trace half-edges in order; each bounded counterclockwise cycle defines one face. Faces are then triangulated by ear clipping for rendering.

Reflection rule

If two faces share crease ℓ , the child is the parent reflected in ℓ . Propagated from a root face outward; if any cycle is inconsistent, the method falls back to per-triangle propagation along a spanning tree.

Kinematic animation

Around crease axis (p, q) , with sign $\sigma(e) \in \{\pm 1\}$ and fold angle θ :

$$T_{\text{child}} = R_{(p,q)}(\sigma(e) \theta) T_{\text{parent}}.$$

Rigid plates connected by hinge creases — no elasticity, no collisions. Near the fully folded state, small offsets along the surface normal keep layer order visible.

Exact vs. heuristic — and conclusion

Exact (in this model)

- Parity condition enforced.
- Kawasaki solved to tolerance.
- Maekawa filters every MV candidate.
- Reflection cycles checked for consistency.

Heuristic / design-driven

- Parity repair is not globally optimal.
- Objective measures drift, not physical energy.
- Bern–Hayes is local; not a full global classifier.
- Interlacing score is aesthetic only.

Delaunay graph → Kawasaki repair → MV search → animated 3D fold.

Thank you.